Vikram Aditya

AI Engineer | Generative AI, LLMs and RAG | Python and Java Backend | Microsoft Azure, CI/CD and MLOps
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SUMMARY

AI Engineer with 2+ years of experience across full-time and internship roles, building production Generative AI and backend systems: Retrieval-Augmented Generation (RAG) pipelines, fine-tuned Large Language Models (LLMs), AI agents with tool calling, and REST APIs deployed on Microsoft Azure with CI/CD and Infrastructure as Code. Primary developer of an internal talent-deployment platform (React, TypeScript, Java Spring Boot, PostgreSQL; 69 REST API endpoints) and owner of the Azure DevOps CI/CD for a second internal platform. Improved retrieval relevance by about 30% across 5+ benchmarked model configurations and cut manual regression testing effort by 60%. Recipient of the ThoughtFocus Most Promising Newcomer award (2025).

CORE SKILLS

Generative AI | Large Language Models (LLMs) | Retrieval-Augmented Generation (RAG) | LLM Fine-Tuning (QLoRA, PEFT) | AI Agents and Tool Calling | Prompt Engineering | Embeddings and Semantic Search | Python | FastAPI | Java | Spring Boot | REST APIs | React | TypeScript | PostgreSQL | Microsoft Azure | Docker | CI/CD | Infrastructure as Code (IaC) | Observability and Monitoring

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL
Generative AI and LLMs: OpenAI API (GPT-4o-mini), Anthropic Claude API, Hugging Face Transformers, PEFT, TRL, Unsloth, QLoRA, LoRA, bitsandbytes, Ollama, Qwen 3, Sentence Transformers, FAISS (vector search), Function Calling, Prompt Engineering
Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Convolutional Neural Networks (CNN), Transfer Learning, TensorFlow, Keras, scikit-learn, NumPy, pandas, OpenCV, NLTK, Stable Diffusion
Backend: FastAPI, Flask, Django, Spring Boot, Spring Security (OAuth2), Spring Data JPA, Node.js, REST APIs, Role-Based Access Control (RBAC), Single Sign-On (SSO), API Authentication, Scheduled Jobs
Frontend: React, TypeScript, Vite, Tailwind CSS
Cloud and DevOps: Microsoft Azure (Container Apps, Azure Container Registry (ACR), PostgreSQL Flexible Server, Blob Storage, Front Door, Web Application Firewall (WAF), Key Vault, Entra ID / Azure AD, Managed Identity), Azure DevOps, Azure Repos, GitHub Actions, Docker, Docker Compose, Bicep (Infrastructure as Code), Google Cloud Platform (GCP)
Observability and Reliability: OpenTelemetry (OTel), Prometheus, Grafana, Jaeger, Alertmanager, Distributed Tracing, Alerting, Rate Limiting, Circuit Breakers, Fault Injection
Databases: PostgreSQL, MySQL, Redis, Flyway
Testing and Tools: Playwright (end-to-end testing), Selenium, pytest, Git, GitHub

PROFESSIONAL EXPERIENCE

ThoughtFocus, Bengaluru, India — Jan 2025 – Present
Associate Engineer, AI and Backend — *Aug 2026 – Present*
Trainee Engineer — *Jan 2025 – Aug 2026*

HR Knowledge Assistant (Generative AI, RAG, LLM fine-tuning)

- Built a Retrieval-Augmented Generation (RAG) assistant that answers employee questions over 200+ HR policy documents, using FAISS vector search, Sentence Transformers embeddings and semantic chunking for policy-grounded answers.
- Fine-tuned a Qwen 3 large language model with QLoRA (Unsloth) and served it through Ollama for domain-specific responses.
- Implemented AI agent tool calling (function calling) that raises IT tickets and sends automated HR notification emails from within the conversation, with role-based access control (RBAC) over document retrieval.

PitStop Connect: talent-deployment platform (primary developer)

- Designed and built the platform end-to-end as primary developer: React and TypeScript frontend, Java Spring Boot backend and PostgreSQL, exposing 69 REST API endpoints across 16 backend modules.
- Integrated LLMs (OpenAI GPT-4o-mini with an Anthropic Claude fallback behind a common interface) to generate hiring rationales for skill-based job-description matching, and extracted resume text from PDF and Word files (PDFBox, Apache POI).
- Automated reservation and extension-approval workflows, a nightly HRMS employee sync and database-driven email notifications with scheduled jobs (Spring Scheduler, ShedLock) and an email queue.

- Secured the platform with Azure AD (Entra ID) single sign-on over OAuth2, role-based access control, API-key authentication for a data-export API, and Azure Front Door with Web Application Firewall (WAF) rate limiting.
- Provisioned dev and prod environments with Infrastructure as Code (Bicep): Azure Container Apps, Azure Container Registry, PostgreSQL Flexible Server and Blob Storage; set up GitHub Actions CI on every pull request, an OIDC-authenticated deployment workflow with versioned Docker images, and Playwright end-to-end tests.

ALIGN: resource-allocation platform (Azure deployment and CI/CD owner)

- Designed the Azure DevOps CI/CD pipeline and multi-stage Docker build: cloud builds in Azure Container Registry, database schema migrations guarded against data loss, health-checked Container App revisions and rollback steps.
- Deployed the platform to Azure Container Apps with passwordless access through a user-assigned managed identity and workload identity federation, Azure Key Vault secrets and Entra ID single sign-on; migrated the codebase to Azure Repos and ran the dev releases.

Quality engineering and team

- Automated regression testing of Adobe Experience Manager (AEM) forms with Selenium and Faker-generated test data, reducing manual regression testing effort by 60%.
- Mentored an intern on retrieval pipeline development, improving delivery quality and implementation speed.

Tech Mahindra Nov 2024 – Jan 2025

AI/ML Engineer Intern, Project Indus (enterprise LLM initiative)

- Built and evaluated RAG workflows and benchmarked 5+ model configurations, improving retrieval relevance by about 30%.
- Ran LLM experiments with Hugging Face Transformers, PEFT, TRL and bitsandbytes on GPU-backed Google Cloud Platform (GCP).
- Built Django and MySQL backend REST APIs for an internship application platform.

Mavencart Jun 2023 – Aug 2023

Backend Developer Intern

- Developed backend services for two generative AI products, an AI writer and a social media template generator, reducing manual content creation effort by about 40%.

PROJECTS

LLM Sentinel: LLM API Gateway (Python, FastAPI, Redis, OpenTelemetry, Docker) github.com/vxkram/llm-sentinel

- Built an LLM API gateway with Redis Lua rate limiting and per-team token budgets, a circuit breaker with automatic provider fallback, and an admin API with hot-reloaded configuration.
- Instrumented it with OpenTelemetry distributed tracing (Jaeger), Prometheus metrics with alert rules (Alertmanager) and Grafana dashboards; verified failover with fault injection in a 9-container Docker Compose stack that runs at $0 via Ollama and OpenAI/Anthropic-compatible mock providers.

Intelligent Tutoring System (Python, FastAPI, React) github.com/vxkram/intelligent-tutoring-system

- Built a full-stack adaptive learning platform that scores quizzes, recommends tiered resources and summarizes PDFs with PyMuPDF and NLTK; found and fixed a scoring bug that capped every quiz at 40%.

Resume Summarizer (Python, Django, React, OpenAI API) github.com/vxkram/resume-summarizer

- Built an LLM document-analysis tool that extracts resume sections and generates summaries and tailored interview questions.

Face Mask Detection (TensorFlow, Keras, MobileNetV2, Flask) github.com/vxkram/face-mask-detection

- Trained a MobileNetV2 convolutional neural network with transfer learning to 97.6% validation accuracy and served it through a Flask API with a web interface.

EDUCATION

Bachelor of Engineering (B.E.), Artificial Intelligence and Machine Learning 2020 – 2024

MVJ College of Engineering, Bengaluru, India

AWARDS

Most Promising Newcomer Award, ThoughtFocus 2025

Presented by the Group Director and Head of HR (India) for outstanding contribution above and beyond the call of duty.